

ROADMAP 2026

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PHASE 0: MINDSET & FOUNDATION **(Week 0)**

Understand **what cybersecurity & pentesting really are** and set expectations.

Core Concepts

- What is **Cybersecurity**
- What is **Penetration Testing**
- Blue Team vs Red Team vs Purple Team
- Legal & ethical boundaries
- Career paths:
 - SOC Analyst
 - **Pentester**
 - **Red Teamer**
 - Threat Hunter
 - AppSec / CloudSec

PHASE 1: IT & COMPUTER BASICS

(Month 1–2)

- ✗ Most beginners fail here.
- ✓ Strong fundamentals = fast growth later.

Operating Systems

Linux (MANDATORY)

- File system
- Permissions
- Users & groups
- Services
- Bash basics

Windows (MANDATORY)

- Users, Groups
- NTFS permissions
- Services
- Registry
- Event logs

Networking (VERY IMPORTANT)

- OSI & TCP/IP models

- IP, Subnetting
- TCP vs UDP
- DNS, HTTP/HTTPS
- Common ports & services
- NAT, Firewall basics

Tools to Practice

- VirtualBox / VMware
 - Kali Linux
 - Ubuntu
 - Windows 10/11
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PHASE 2: BASIC PROGRAMMING (Month 3)

Why?

You don't need to be a **developer**, but you must be able to **read, modify, and write scripts**.

Languages (Priority Order) → Not Mandatory for beginners.

1. **Python** ★
2. Bash
3. Basic JavaScript (for web attacks)

Pentest-Relevant Skills

- Modify exploit scripts
- Write automation scripts
- Parse outputs
- API testing

PHASE 3: WEB FUNDAMENTALS (Month 4)

Most vulnerabilities live here.

Web Basics

- HTML / CSS (reading level)
- JavaScript (logic level)
- HTTP requests & responses
- Cookies & sessions
- Authentication vs Authorization

Backend Basics

- PHP / Node / Python (conceptual)
- APIs (REST)
- JSON / XML

Deliverables

- ✓ Understand how a web app works internally
 - ✓ Can intercept & analyze traffic
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PHASE 4: WEB SECURITY & OWASP (Month 5–6)

Core Vulnerabilities (MANDATORY)

- SQL Injection
- XSS (Stored / Reflected / DOM)
- IDOR
- Authentication bypass
- File upload vulnerabilities
- Command Injection
- CSRF
- SSRF
- Business Logic Flaws

Standards

- OWASP Top 10 (2021 → 2025 concepts)
- OWASP ASVS (basic idea)

Tools

- Burp Suite

- sqlmap
- ffuf / wfuzz
- nikto

Deliverables

- ✓ Exploit OWASP Top 10 manually
 - ✓ Write vulnerability reports
 - ✓ Understand impact & remediation
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PHASE 5: SYSTEM & NETWORK PENTESTING (Month 7–8)

Linux Attacks

- Enumeration
- SUID abuse
- Cron jobs
- Misconfigurations
- Kernel exploits (conceptual)

Windows Attacks

- Privilege escalation
- Services abuse
- Credential dumping (concept)
- Scheduled tasks

Networking Attacks

- SMB
- FTP
- SSH
- SNMP
- NFS

Tools

- nmap
- enum4linux
- linpeas / winpeas
- netcat

Deliverables

- ✓ Root Linux machines
 - ✓ Admin Windows machines
 - ✓ Explain attack chains
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PHASE 6: ACTIVE DIRECTORY (Month 9)

This is where professionals are separated from beginners.

Core Concepts

- Domain, Forest, Trust
- Kerberos

- LDAP
- GPO

Attacks

- Kerberoasting
- AS-REP Roasting
- Pass-the-Hash
- Pass-the-Ticket
- ACL abuse
- BloodHound analysis

Tools

- BloodHound
- SharpHound
- impacket
- Rubeus

Deliverables

- ✓ Domain compromise
- ✓ Attack path analysis
- ✓ Lateral movement mastery

PHASE 7: CLOUD & MODERN TECH **(Month 10)**

Cloud Basics

- AWS / Azure fundamentals
- IAM misconfigurations
- Cloud storage exposure
- Token abuse

Modern Security

- APIs security
- OAuth / OIDC basics
- JWT attacks
- CI/CD risks

Deliverables

- ✓ Understand cloud attack surface
 - ✓ Test APIs confidently
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PHASE 8: REPORTING & PROFESSIONAL SKILLS (Month 11)

Reporting (VERY IMPORTANT)

- Executive summary
- Risk rating
- Reproduction steps
- Impact

- Remediation

Soft Skills

- Client communication
- Scoping
- Ethics
- Legal authorization
- Time management

Deliverables

- ✓ Client-ready pentest reports
 - ✓ Confident communication
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PHASE 9: CERTIFICATION & JOB READINESS (Month 12)

Entry → Advanced

- eJPT → PNPT → OSCP
- Bug bounty basics
- Freelance readiness
- Interview prep

Portfolio

- Writeups (sanitized)
- GitHub scripts

- Blog / LinkedIn presence

PHASE 0: CYBERSECURITY INTRO & MINDSET

- **Cybrary**
<https://www.cybrary.it>
 - **Open Security Training**
<https://opensecuritytraining.info>
 - **OWASP**
<https://owasp.org>
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PHASE 1: LINUX, WINDOWS & NETWORKING

Linux

- **Linux Journey**
<https://linuxjourney.com>
- **OverTheWire**
<https://overthewire.org/wargames/>

Windows

- **Microsoft Learn**
<https://learn.microsoft.com>

Networking

- **Professor Messer**
<https://www.professormesser.com>
 - **Cisco Networking Academy**
<https://www.netacad.com>
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PHASE 2: PROGRAMMING (PYTHON / BASH / JS)

- **freeCodeCamp**
<https://www.freecodecamp.org>

- **Automate the Boring Stuff with Python**
<https://automatetheboringstuff.com>
 - **GNU Bash Manual**
<https://www.gnu.org/software/bash/manual/>
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PHASE 3: WEB FUNDAMENTALS

- **MDNouting Web Docs**
<https://developer.mozilla.org>
 - **PortSwigger Web Security Academy**
<https://portswigger.net/web-security>
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PHASE 4: WEB SECURITY & OWASP TOP 10

- **OWASP WebGoat**
<https://owasp.org/www-project-webgoat/>
 - **OWASP Juice Shop**
<https://owasp.org/www-project-juice-shop/>
 - **Hack This Site**
<https://www.hackthissite.org>
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PHASE 5: SYSTEM & NETWORK PENTESTING

- **TryHackMe**
<https://tryhackme.com>
- **VulnHub**
<https://www.vulnhub.com>
- **Hack The Box**
<https://www.hackthebox.com>
- **GTFOBins**
<https://gtfobins.github.io>
- **LOLBAS**
<https://lolbas-project.github.io>

PHASE 6: ACTIVE DIRECTORY

- **BloodHound**
<https://github.com/BloodHoundAD/BloodHound>
- **impacket**
<https://github.com/fortra/impacket>
- **HackTricks**
<https://book.hacktricks.xyz>
- **IppSec**
<https://www.youtube.com/@ippsec>

PHASE 7: CLOUD & API SECURITY

- **AWS Skill Builder**
<https://skillbuilder.aws>
- **Cloud Security Alliance**
<https://cloudsecurityalliance.org>
- **OWASP API Security Top 10**
<https://owasp.org/www-project-api-security/>

PHASE 8: REPORTING & BUG BOUNTY LEARNING

- **HackerOne**
<https://hackerone.com/hackactivity>
- **Bugcrowd**
<https://bugcrowd.com>

PHASE 9: COMMUNITY & CONTINUOUS LEARNING

- **Reddit**
<https://www.reddit.com/r/netsec>
<https://www.reddit.com/r/oscp>
- **GitHub**
<https://github.com>

- **Medium**
<https://medium.com>
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RECOMMENDED

- <https://portswigger.net/web-security>
- <https://tryhackme.com>
- <https://book.hacktricks.xyz>
- <https://www.vulnhub.com>
- <https://overthewire.org>

HIGH-VALUE CYBERSECURITY CHANNELS (ADD LINKS)

Here are core channels you should **link to in your PDF** (great for beginners → advanced):

10. HackerSploit (Penetration Testing & Offensive Security)

<https://www.youtube.com/channel/UC0ZTPkdxIAKf-V33tqXwi3Q> [YouTube](#)

11. NetworkChuck (Linux / Security Basics)

<https://www.youtube.com/networkchuck> [YouTube](#)

12. John Hammond (In-Depth Hacking Tutorials)

<https://www.youtube.com/c/JohnHammond010> [APLYiD](#)

13. David Bombal (Networking + Cybersecurity)

https://www.youtube.com/channel/UCP7WmQ_U4GB3K51Od9QvM0w [YouTube](#)

14. IppSec (HTB Walkthroughs – Pentesting Practice)

<https://www.youtube.com/@ippsec> [APLYiD](#)

15. LiveOverflow (Hacking & CTF / Vulnerabilities)

 <https://www.youtube.com/c/LiveOverflow>

16. WsCube Cyber Security (Bangla tutorials)

 <https://www.youtube.com/@WsCubeCyberSecurity>

Online Platforms and Labs

These are hands-on environments where beginners can practice ethical hacking safely and legally.

- TryHackMe: Gamified learning with beginner rooms on topics like reconnaissance, scanning, and web vulnerabilities. Free tier available with guided paths.[reddit.com](#)
- Hack The Box (HTB): Offers free academy modules for beginners, including challenges on penetration testing. Great for practical skills like exploiting machines.[reddit.com](#)
- PortSwigger Web Security Academy: Free labs focused on web hacking, including SQL injection, XSS, and more. Ideal for bug bounty hunters.[reddit.com](#)[cybrvault.com](#)
- VulnHub: Downloadable virtual machines for offline pentesting practice. Free and community-driven.[@trickyhash](#)
- DVWA (Damn Vulnerable Web Application): A free web app for practicing common vulnerabilities like SQL injection.[@trickyhash](#)
- WebGoat: Java-based vulnerable app for learning web security flaws.[@trickyhash](#)
- bWAPP (buggy web application): PHP-based platform for testing over 100 vulnerabilities.[@trickyhash](#)
- OWASP Juice Shop: A modern web app with vulnerabilities for pentesting practice.[@trickyhash](#)
- CTFtime.org: Directory of free Capture The Flag (CTF) events for skill-building.

Free Courses and Tutorials

Structured learning paths to build foundational knowledge.

- Cisco Networking Academy Ethical Hacker Course: Free intro course with CTF challenges. Covers basics like reconnaissance and scanning.[netacad.com](#)
- EC-Council Ethical Hacking Essentials: Free essentials bundle for beginners.[@DS_Watch](#)
- ZSecurity Learn Ethical Hacking From Scratch: Comprehensive course on basics to advanced topics. Available via links shared in communities.[@TheSecMaster1](#)[@thesaurabh](#)
- TCM Security Practical Ethical Hacking: 20-hour hands-on course focusing on practical skills over theory.[@TCMSecurity](#)
- Udemy Free Courses: Options like "The Complete Ethical Hacker Developer Course - Introduction" (1.9 hours) and "Learn Coding Basics for Ethical Hacking" (1.9 hours).[@comidoc@comidoc](#)
- freeCodeCamp: Python and networking basics, essential for scripting in hacking.
- Khan Academy: Free networking fundamentals.

Books and PDFs

Downloadable resources for self-study (ensure legal sources).

- Ethical Hacking and Penetration Testing Guide: Covers Kali Linux and pentesting techniques. [@DIGITALEARN_1](#)
- Beginning Ethical Hacking with Kali Linux: Beginner-friendly guide to tools and commands. [@DIGITALEARN_1@DIGITALEARN_1](#)
- A Beginner's Guide to Ethical Hacking with Kali & Cybersecurity: Includes Linux command line and security tools. [@DIGITALEARN_1](#)
- Hacking: The Art of Exploitation: Classic book on low-level hacking.
- The Web Application Hacker's Handbook: Focuses on web vulnerabilities.
- Mega Folders of Resources: Community-shared collections like 51.72 GB of cybersecurity materials, including ethical hacking guides. [@bountywriteups@NitinGavhane](#)
- Free Ethical Hacking Course PDF: Structured guide covering lab setup, recon, scanning, web pentesting, DoS, malware, social engineering, cryptography, and reporting. [@TheMsterDoctor1](#)
- Learn Ethical Hacking from Scratch PDF: Guide on hacking computers ethically [@_0b1d1](#)

Certifications

Entry-level certs to validate skills (some with free prep materials).

- eJPT (eLearnSecurity Junior Penetration Tester): Hands-on for beginners.
- CompTIA Security+: Foundational cybersecurity cert.
- CEH (Certified Ethical Hacker): Covers ethical hacking theory.
- OSCP (Offensive Security Certified Professional): Advanced, but prep with free resources.